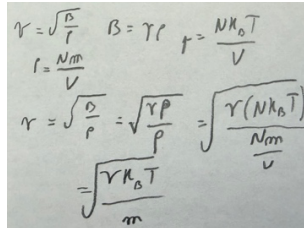


Lab 6: Speed of Sound

Pre Lab

Activity 1

Exercise 1: Solve for the speed of sound



$$\gamma = \frac{C_p}{C_v} \quad B = \gamma P \quad f = \frac{Nk_B T}{V}$$

$$f = \frac{Nm}{V}$$

$$\gamma = \sqrt{\frac{C_p}{C_v}} = \sqrt{\frac{\gamma P}{P}} = \sqrt{\frac{\gamma (Nk_B T)}{\frac{Nm}{V}}}$$

$$= \sqrt{\frac{\gamma Nk_B T}{m}}$$

Exercise 2: What mass will we use

- 4.81×10^{-26} kg, this is the value of one molecule of dry air extrapolated from NASA's published molar mass of dry air divided by Avogadro's number.
- $m = m_{\text{air}}/N_a \rightarrow N_a = 6.022 \times 10^{23}$ |||| m_{air} is given by [NASA](#)
- $v(T) = 20.0 \sqrt{T} \frac{m}{s}$

Exercise 3: What are the model's assumptions

- Air behaves as an ideal gas
- The composition of the air is constant
- Temperature is constant
- The process is adiabatic, not isothermal

Exercise 4: What experiment will we do

- We will determine the speed of sound using a speaker and a microphone inside a one-end open tube. We will use the change in the offset between the function generator's output and the microphone's input at different distances along the tube. We will drive the speaker with a square wave. Split the generator output to the oscilloscope CH1 and to the speaker. Send the microphone signal to CH2 via the amplifier. Move the microphone to several known positions inside the tube. For each position, record the distance x and the delay t between CH1 and CH2.
- Plot t on the vertical axis versus x on the horizontal axis. Use at least 6 to 10 positions spanning the usable tube length $\rightarrow t = t_0 + \frac{x}{v}$
- The slope is $\frac{1}{v} \rightarrow$ Its reciprocal gives the speed of sound v

- d. The intercept t_0 is the fixed time offset.

Exercise 5: Procedure for the experiment

1. Plug in the microphone to the oscilloscope as well as the function generator
 2. Set the function generator to square wave and find a good frequency for our lab
 - a. Keep to low frequencies + make sure wavelength is workable with a strong visible peak
 3. Place the microphone inside the tube and position the speaker at the other end.
 4. Measure the starting distance between the microphone and the speaker
 5. Set up the data tables used for this lab
 6. Take 15-20 datapoints
 - a. For each datapoint, measure the distance from the speaker to the microphone
 - b. Calculate the offset between the edge of the step function and the nearest input peak from the microphone
 - c. Plot point on datasheet
 7. Use the cursor function on the oscilloscope to measure each datapoint for accuracy.
 8. Plot t vs. x and perform a linear fit
 9. Answer all question in lab manual
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Actual Lab

Exercise 6: Frequency

- a. What frequency will you use for your experiment?
 - a. I am using a **486.4 Hz $\pm$ 0.5 Hz** square wave
- b. Describe what you did to determine this.
 - a. I tested several frequencies in the lower audio range. At **486.4 Hz $\pm$ 0.5 Hz**, the microphone picked up strong, well-defined peaks, and the step function looked well-spaced. Lower frequencies produced long periods, reducing precision in the timings. Higher frequencies caused the pulses to overlap, resulting in a less consistent microphone response.
- c. Are there any issues you can identify with setting the frequency too high or too low?
 - a. Too low: The period becomes too long, reducing the precision of the delay measurement on the oscilloscope.
 - b. Too high: The microphone output becomes less clear, and reflections within the tube begin to overlap.

Exercise 7: How to handle the delay

- a. We can handle the delay by absorbing it into the y-intercept of the linear fit ($t = t_0 + \frac{x}{v}$). When plotting t versus x , the delay automatically becomes the intercept, since it is constant for all values, and the slope of the equation is unaffected, thereby preserving the accuracy of the speed of sound, even given an unknown time delay from our equipment.

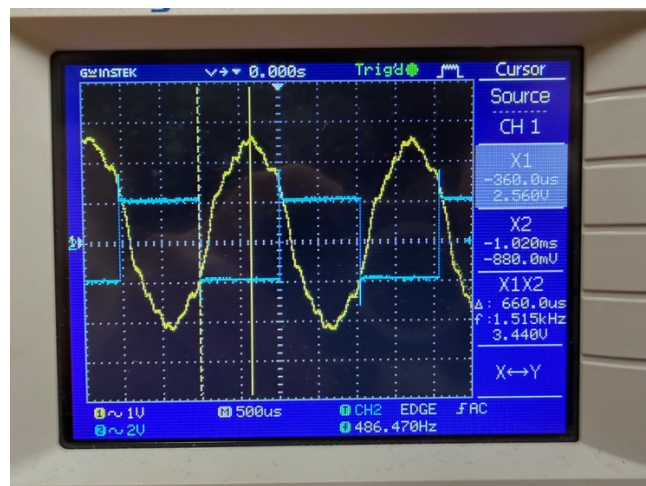
Activity 2

Exercise 8: Notes throughout the exercise

- a. The whole procedure went smoothly. One small mistake I made at first was when gathering the data, I accidentally messed with the wave generator and changed the wave by 1Hz throughout the experiment and had to redo those points on my plot. After retesting the messed-up points, my data became more accurate, and my uncertainty went down. The oscilloscope traces were clean, with a good, clear peak for me to record. The step function was clearly defined as well. The uncertainty of my measured output was a little high at $1 \cdot 10^1$ m/s. My temperature number is also likely incorrect, as I had to use the value of the outdoor weather and not an accurate lab measurement of the current temperature. Because my data is very close to the expected value, this temperature offset likely had an impact, creating a false positive result.

Exercise 9: The oscilloscope with a microphone

- a. Take a picture of your oscilloscope screen while the microphone is picking up a signal and include it here.



b. Describe what you are measuring.

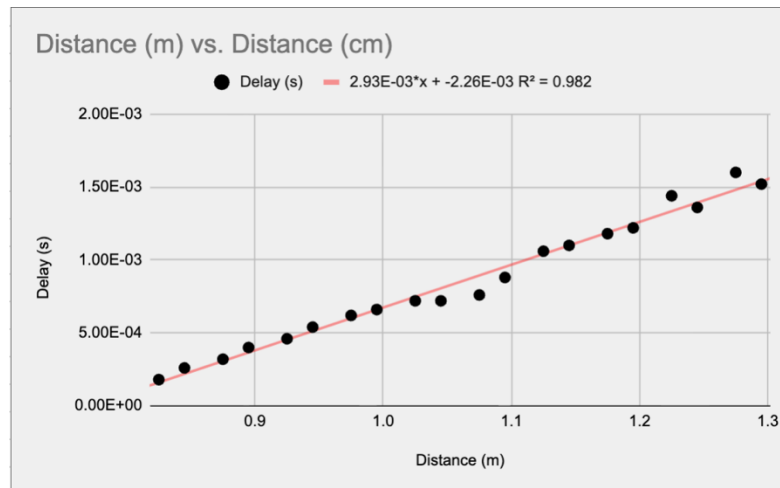
- a. We are measuring the phase difference between the oscilloscope's output (through the speaker) and the microphone input. This shows up as the time delay between the function generator signal (CH1) and the microphone signal (CH2). We now have a time for each offset (t) for each distance (x) that we test. We will use this relationship to determine the speed of sound in our lab environment.

Exercise 10: Data tables

Trial Number	Distance (cm)	Distance (m)	Delay (ms)	Delay (s)		Frequency:	486.4	Hz
1	102.5	1.025	0.720	7.20E-04				
2	97.5	0.975	0.620	6.20E-04		Room temp:	15	C
3	92.5	0.925	0.460	4.60E-04			289.15	K
4	87.5	0.875	0.320	3.20E-04				
5	82.5	0.825	0.180	1.80E-04		Uncertainty	0.5	mm
6	107.5	1.075	0.760	7.60E-04			0.0005	m
7	112.5	1.125	1.060	1.06E-03			0.5	Hz
8	117.5	1.175	1.180	1.18E-03			0.04	ms
9	122.5	1.225	1.440	1.44E-03			4.00E-05	s
10	127.5	1.275	1.600	1.60E-03			0.5	C
11	99.5	0.995	0.660	6.60E-04				
12	94.5	0.945	0.540	5.40E-04				
13	89.5	0.895	0.400	4.00E-04				Uncertainty
14	84.5	0.845	0.260	2.60E-04		speed of sound:	341	1E+01
15	104.5	1.045	0.720	7.20E-04		Slope:	0.00293	9E-05
16	109.5	1.095	0.880	8.80E-04		X intercept:	0.770	4E-02
17	114.5	1.145	1.100	1.10E-03		Y intercept:	-0.00226	1E-04
18	119.5	1.195	1.220	1.22E-03				
19	124.5	1.245	1.360	1.36E-03		Model speed of sound:	340.0882238	0.2940413486
20	129.5	1.295	1.520	1.52E-03				

Activity 3

Exercise 11: The plot for this exercise



Exercise 12: Analysis

a. What is the slope of your line of best fit?

a. The slope for the line of best fit is:

$$m = 0.00293 \pm 9 \cdot 10^{-5} \frac{s}{m}$$

b. What are the x and y intercepts?

a. The x-intercept is:

$$X_{\text{int}} = 0.770 \pm 4 \cdot 10^{-2} \text{ m}$$

b. The y-intercept is:

$$b = -0.00226 \pm 1 \cdot 10^{-4} \text{ s}$$

Exercise 13: Speed of sound conclusions

a. What is your measured speed of sound in air? Include uncertainty.

a. The measured speed of sound in air is:

$$v_{\text{meas}} = 341. \pm 1 \cdot 10^1 \frac{m}{s}$$

b. How does this compare to the speed of sound predicted by your model? Include uncertainty.

a. The model predicts the speed of sound to be:

$$v_{\text{model}} = 340. \pm 0.3 \frac{m}{s}$$

The model's predicted value is within 1 m/s of the measured value, and the range of the measured value is 331 m/s to 351 m/s, which covers the range of our expected value of 339.7 m/s to 340.3 m/s. The uncertainty of our measured value is substantial at $1 \cdot 10^1$. This means our data could benefit from more data points. The R^2 value for our dataset was 0.982, indicating that our linear approximation is a good fit to the data.

Exercise 14: If your measurement is in agreement with theory, how would you improve this experiment to increase your precision to be sure?

- a. My measurement agrees with theory; thus, some simple improvements to this system would be to gather the distance data using more precise tools. I had to eyeball the tube entrance using the marks on the ruler, which introduced error. Using lasers to mark the tube entrance and determine its length would be very helpful, as would a more precise ruler/measuring tool. A microphone more closely calibrated for this test could help, as the input it received seemed distorted (a better microphone might fix some but not all of it). I could spend more time finding a cleaner frequency to achieve a cleaner peak for testing. Testing in a well soundproofed room where the air density is measured, and the temperature is known and constant. Not having access to a thermometer inside the lab reduces the accuracy of my results, as I have to use a locationally averaged temperature report (Isla Vista weather) rather than one specific to the lab room I'm in.